

2024



crypto.com

Crypto.com Delivers Accurate Sentiment Analysis in 1 Second with Generative AI on AWS

Crypto.com uses Amazon Bedrock with Amazon SageMaker Studio to run an efficient architecture that delivers nuanced, domain-specific crypto market insights to 100 million global users.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

1 second

to return results from large language models

25 languages

Delivers localized multilingual content



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



I



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



Rapid testing

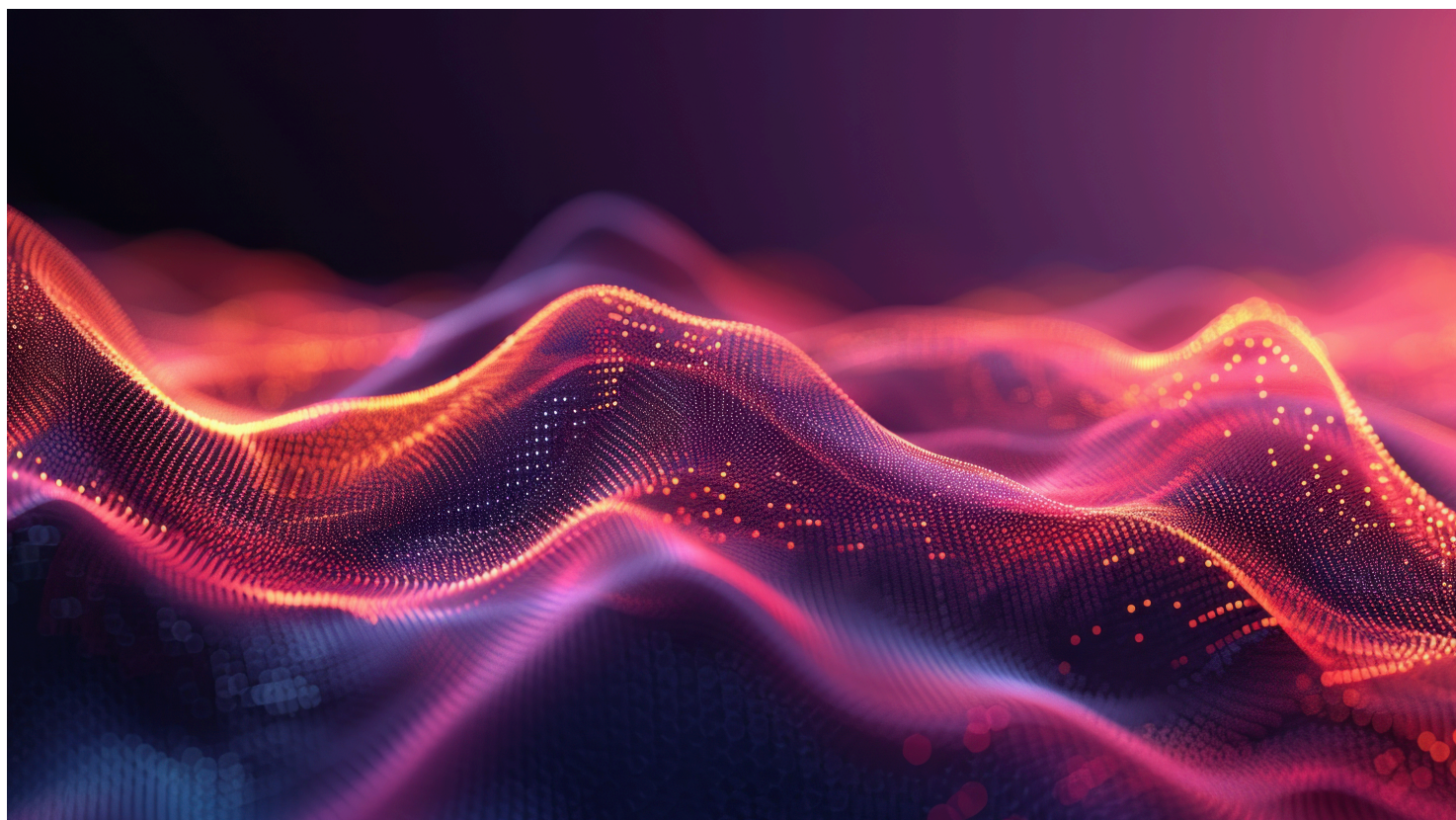
Helps engineers continually test new models



Overview

[Crypto.com](#) is a crypto exchange and comprehensive trading platform serving 100 million users in 90 countries. To improve the service quality of Crypto.com, the firm implemented generative artificial intelligence (AI)-powered sentiment analysis services on AWS to generate market insights.

Crypto.com uses [Anthropic Claude 3](#) large language models on [Amazon Bedrock](#) for sentiment analysis and application development, and [Amazon SageMaker](#) to fine-tune its custom models. With the top models on AWS, the company can deliver accurate sentiment analysis in less than 1 second, and efficiently train and adjust new models in a rapidly evolving market.



Opportunity | Fostering Crypto Adoption with 100 Million Global Users

[Crypto.com](#) was founded in 2016 with a bold mission: to bring cryptocurrency to every wallet. The company distinguishes itself from other trading platforms through its strong focus on driving user adoption, with a wide network of partners including merchants and payment gateways. Crypto.com currently serves around 100 million users in 90 countries.

With such rapid growth and a diverse customer base—all within an evolving, competitive industry—Crypto.com has embraced generative artificial intelligence (AI) to quickly deliver optimized customer experiences. The company's AI use cases include conversational assistants for onboarding and customer queries, plus wizards to generate social media marketing campaigns.

In generative AI applications, sentiment analysis and news narrative categorization are becoming increasingly vital; particularly in the cryptocurrency space, investors need timely and accurate market intelligence to inform their decisions. Crypto.com provides a market insight service where users can access the latest news and analysis from both crypto and traditional news sources to gain insights from the company's intelligent subscription is tailored to the user's trading level and the coins in their wallet.

Crypto.com employs a range of off-the-shelf machine learning (ML) models alongside its own custom models for sentiment analysis. However, developers encountered challenges with the limitations of open-source models and the high cost of self-hosting large language models (LLMs). They also faced accuracy issues with outputs generated by open-source models, especially when dealing with multilingual news sites. Consequently, Crypto.com sought a more effective solution to integrate and synthesize outputs from multiple ML models—both pre-trained and custom—to provide accurate, reliable, and comprehensive insights into the crypto market.

Solution | Facilitating Development with Off-the-Shelf and Custom Models

Crypto.com has run on Amazon Web Services (AWS) since its launch, so when it sought new LLMs for sentiment analysis, [Anthropic Claude 3](#) on [Amazon Bedrock](#) was a logical choice. These LLMs are highly scalable and process vast amounts of data in real-time, facilitating comprehensive market research. Initial results showed that LLMs on Amazon Bedrock returned results very quickly, typically within one second.

Integrating Amazon Bedrock eliminated the manual effort, extra cost, and computational constraints of self-hosting LLMs. Within one month, Crypto.com had implemented Anthropic Claude 3 Haiku models on Amazon Bedrock for sentiment analysis, collecting and analyzing crypto news in more than 25 languages. The firm also utilizes Amazon Bedrock for ongoing proofs of concept (POCs) and development. Sunny Fok, head of AI & innovation technology at Crypto.com, says, “Generative AI on AWS services like Amazon SageMaker and Amazon Bedrock streamlined our adoption of the latest LLMs and AI technologies. We can now take innovative ideas from POC to full-scale production in weeks.”

To ensure domain-specific knowledge is applied in model output, Crypto.com used its own data to fine-tune open-source models including [Mistral AI](#) and [Meta Llama](#) on [Amazon Elastic Compute Cloud \(Amazon EC2\)](#). This approach is crucial when a new coin appears on the market, as off-the-shelf models often deliver subpar results. Crypto.com then began using [Amazon SageMaker](#) as an on-demand, end-to-end ML development platform to fine-tune its custom models.

Raymond Lam, senior engineer at Crypto.com, says, “Amazon SageMaker provides the tools and infrastructure to adapt our models with a user-friendly interface. Similar to Amazon Bedrock, we only need to run machine learning jobs as



needed, and it's much easier to manage custom modeling than if we were doing it on our own."

Crypto.com received extensive support from AWS throughout the idea generation, POC, and production phases of multi-agent deployment. "We had several discussions on what tools or frameworks to deploy for different use cases," says Lam. "The AWS team shared existing use cases, provided sample code, and demonstrated each step. When technical issues came up, AWS solutions architects helped troubleshoot and offered suggestions, which sped up our onboarding to generative AI on AWS services."

Outcome | Delivering More Comprehensive, Localized Market Insights

By implementing a multi-agent consensus-seeking solution for sentiment analysis on AWS, Crypto.com can efficiently deliver accurate, comprehensive, and localized crypto market insights to its global user base. Fok explains, "We can share more instant, updated news with users on the sentiment of coins—whether they're bullish or bearish, for example. Users become more informed, which means they can make better-planned investments." Engineers have also noted improved accuracy for large-context QE with Claude 3 models on Amazon Bedrock.

Furthermore, with Amazon Bedrock, Crypto.com benefits from highly scalable models that automate insight generation to save time and resources. "We're becoming more efficient with ready-to-use models we can access via API, which gives us more flexibility in development," Lam says. "With generative AI on AWS, we have more options for different use cases or requirements, so we can easily test new models as they come out."

Crypto.com is currently exploring new use cases for Claude 3 models on Amazon Bedrock, such as processing documents, tables, and charts. Initial tests have shown promising results in terms of accuracy compared to ML-driven optical character recognition (OCR) readers on the market. The company is also actively developing new generative AI use cases, such as capturing sentiment in social media.

Customers have expressed satisfaction with the work underway, which encourages Crypto.com to keep testing ways to deploy generative AI across the organization. "The feedback from our generative AI projects has been quite amazing, both internally and from our users," Fok shares.

About Crypto.com

Crypto.com is on a mission to accelerate the world's transition to cryptocurrency. With approximately 100 million users across 90 countries, the Singapore-based company offers a trading platform, derivative exchanges, and more. Crypto.com also partners with merchants and payment gateways to support the development of a global crypto network.

AWS Services Used

Amazon Bedrock



Amazon Bedrock is a fully managed service that offers a choice of high-performing foundation models (FMs) from leading AI companies like AI21 Labs, Anthropic, Cohere, Meta, Mistral AI, Stability AI, and Amazon through a single API, along with a broad set of capabilities you need to build generative AI applications with security, privacy, and responsible AI.

[Learn more »](#)

Amazon SageMaker

Amazon SageMaker JumpStart helps you quickly and easily get started with machine learning.

[Learn more »](#)

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

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1 2 3 4 5 6 ... 8

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